

Lab 8 Experiments using LSTM on IMDB Dataset

Aim:

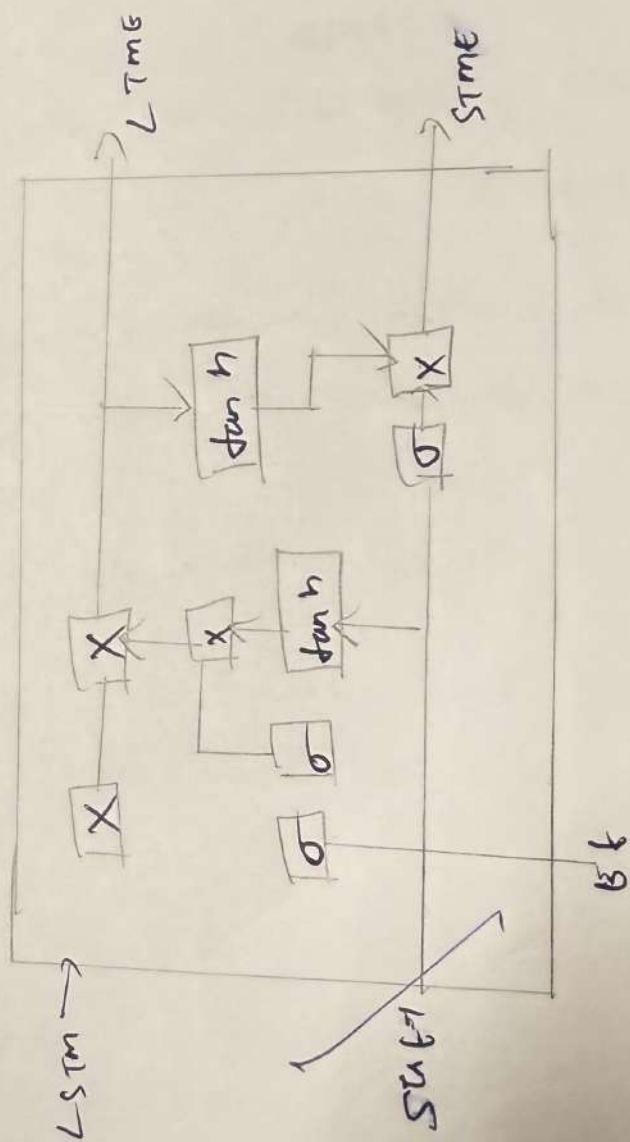
To implement and train a long short-term memory model for sentiment classification

Objective:

1. To understand the arch and working of LSTM networks.
2. To preprocess textual data for recurrent models.
3. To train an LSTM-based model for sentiment.
4. To evaluate the performance of the LSTM on unseen text data.

Pseudocode:

1. Import necessary libraries (TensorFlow/Keras)
2. Load the IMDB data using keras.datasets.imdb.
3. Preprocess data:
 - limit vocab size
 - pad sequence to equal length
4. Build LSTM model.
5. compile model:
6. Train model with training data.
7. plot training accuracy and loss.



Observation:

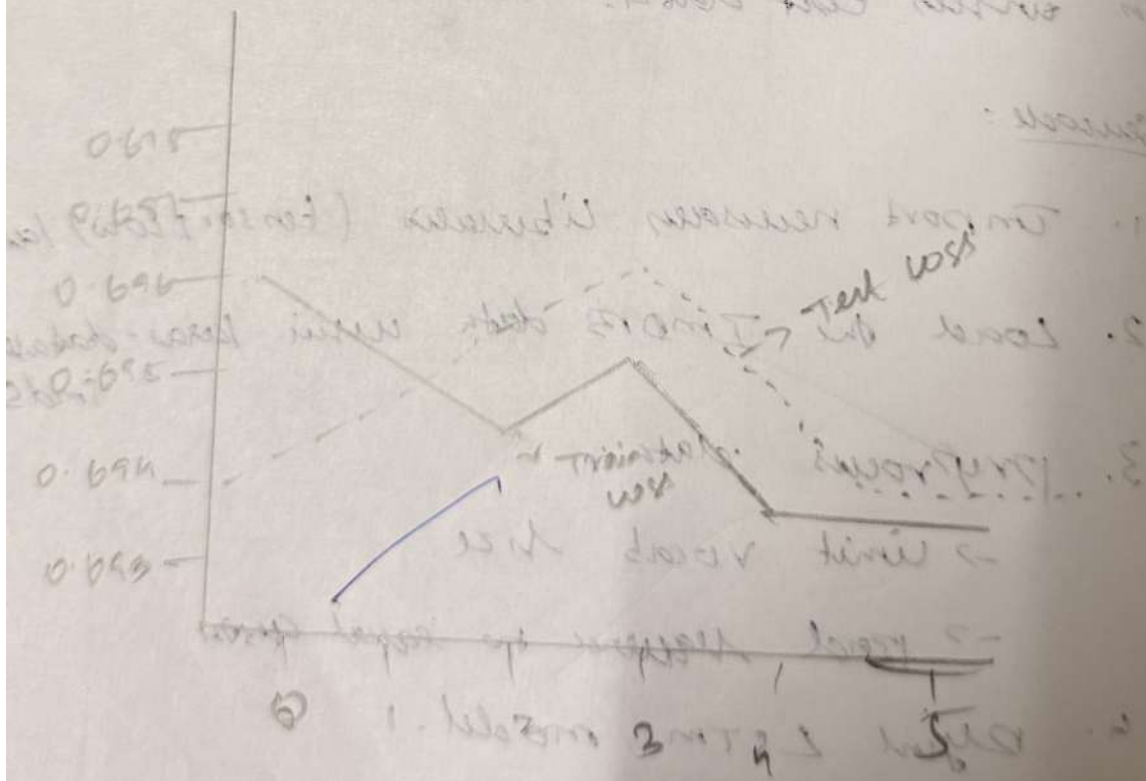
1. Data preprocessing: Text reviews were converted into integer sequence and padded to equal uniform length.
2. Training Behaviour:
 - > Training accuracy steadily improved with epoch, showing good LSTM.
3. Performance: The model achieved high accuracy (85-95%).
4. Visualization: Accuracy and loss curves showed a smooth convergence pattern.

Result:

Successfully implemented an LSTM model.

epoch:		
epoch 1/5	Train loss: 0.6432	Test loss: 0.652
epoch 2/5	Train loss: 0.6435	Test loss: 0.633
epoch 3/5	Train loss: 0.6021	Test loss: 0.690
epoch 4/5	Train loss: 0.6632	Test loss: 0.691
epoch 5/5	Train loss: 0.6013	Test loss: 0.670

Train/Test loss graph: To visualize the performance of the model on training and test data.



Plot training and test loss. Train model using training data. Sample model: 1. loss 2. loss 3. loss 4. loss 5. loss